Neural-Network Automated Chondrule and Components Segmentation (NACHOS)

Towards an Automated Chondrite Classification Method

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Introduction

Chondrites make up 87% of stony meteorites and record the chemical, textural and isotopic diversity of the protoplanetary disk in their components: **chondrules**, refractory inclusions (**CAIs, AOAs**), **matrix** and **metallic phases**. Classification criteria remain largely empirical, quantifying component geometry at scale could sharpen them at minimal cost.

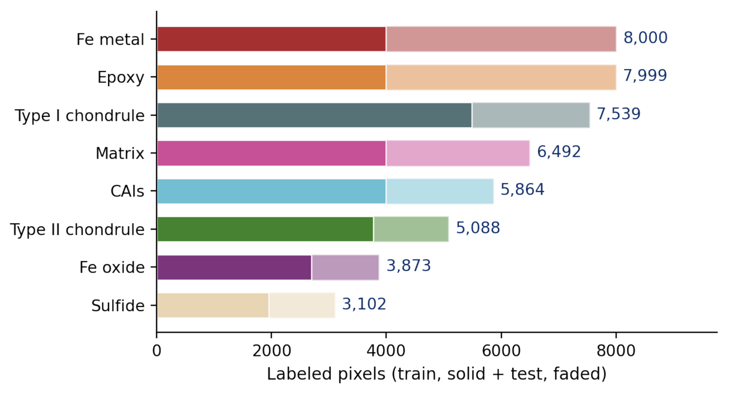
State of the Art

Chondrite classification relies on **manual point counting** under the microscope: a petrologist identifies and measures each component by eye, a slow and subjective process that limits how many sections can be studied. Automated attempts exist, **texture based segmentation of X-ray tomography volumes** [2], but they rely on a single imaging modality and ignore elemental composition, so chemically distinct but texturally similar phases stay unresolved. **NACHOS** instead combines BSE texture with EDS elemental maps.

Data

High resolution (<1 μm/px) BSE and EDS (15 keV) mosaics of **DOM 08006** (CO3.0, pristine), acquired on a TESCAN CLARA: BSE plus 7 elemental maps (Ca, Al, Mg, Si, Fe, S, O). Ground truth comes from a **single expert** via Ilastik point and region annotation. A spatial 70/30 block split (50×50 px tiles) keeps train and test pixels from the same grain apart, giving **47,957 labeled px** across 8 classes. Class colors:

- Background
- Type I chondrule
- Type II chondrule
- Matrix
- CAIs
- Sulfide
- Fe oxide
- Fe metal
- Epoxy
- Boundaries



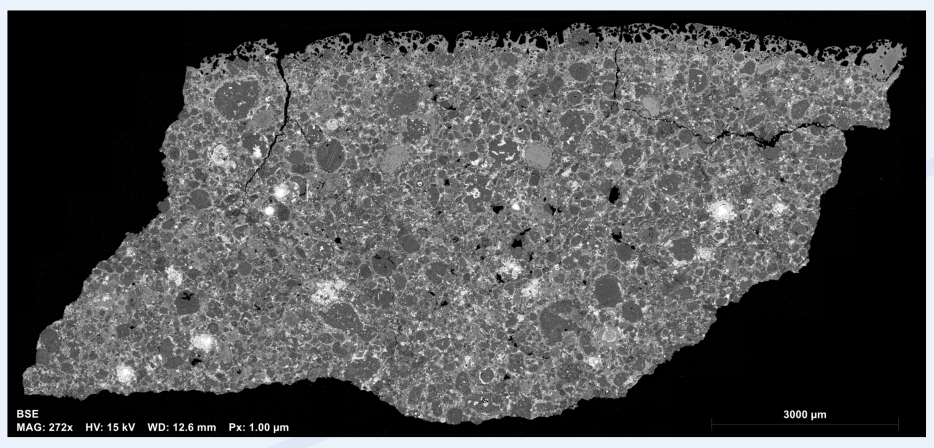


Figure: Full BSE mosaic of DOM 08006.

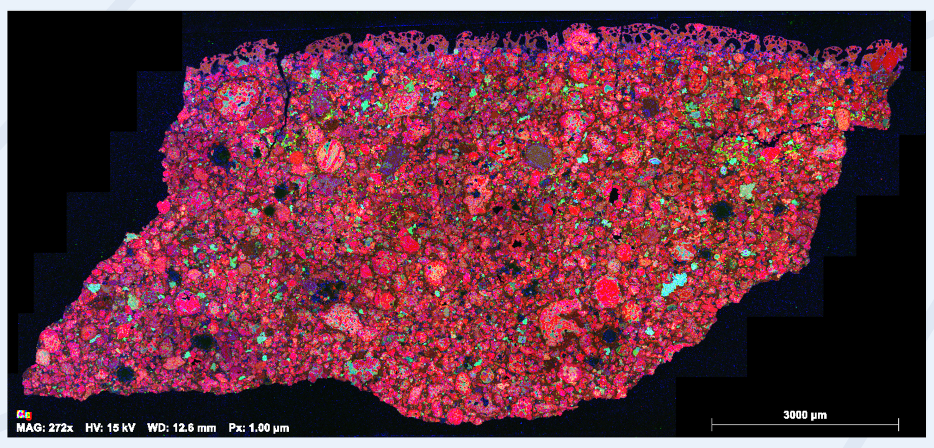
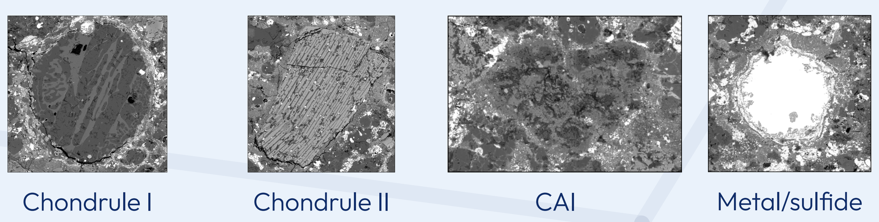
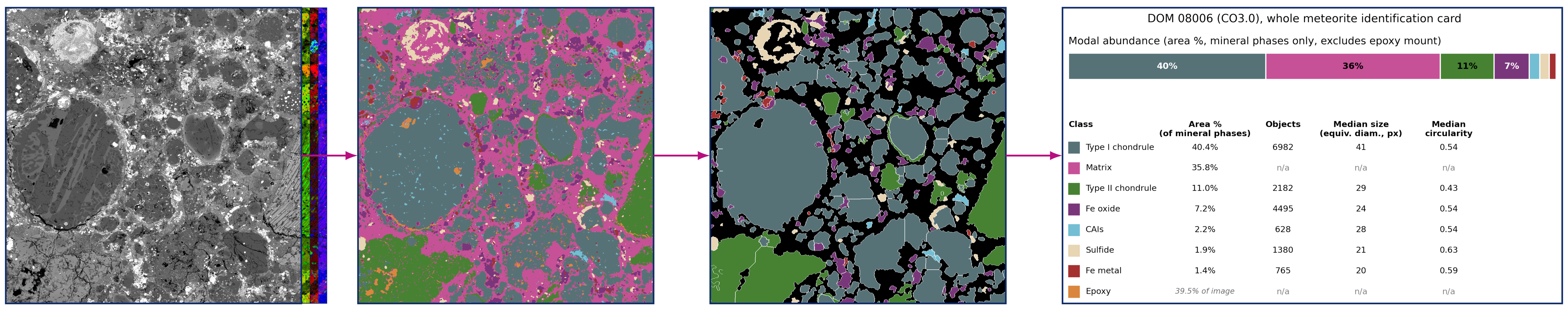


Figure: EDS composite (R = Mg, G = Ca, B = Al).



Method: from Annotated Pixels to Measured Objects

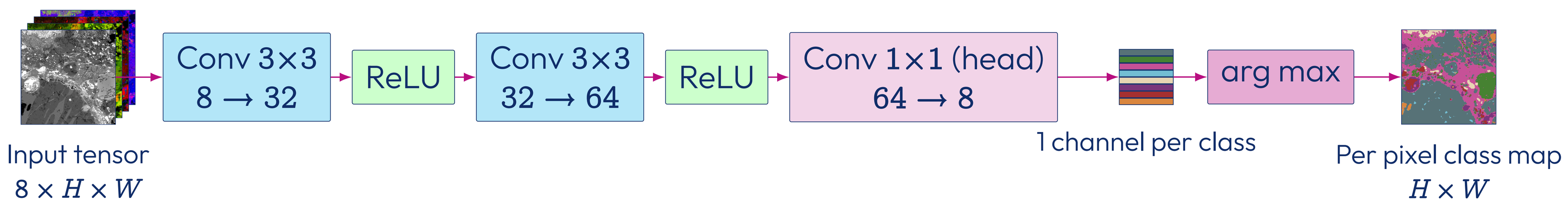
The pipeline turns raw elemental maps into per object measurements in four steps, shown below on one representative region of DOM 08006:



1. Input data Eight channel patches (BSE plus 7 EDS maps) centered on each pixel; dataset size, class list and split are detailed in **Data**, left.

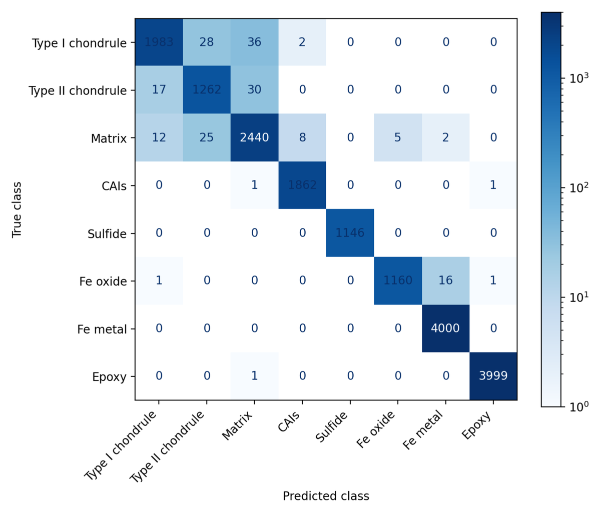
2. Convolutional network

Our model is a custom, lightweight, **fully convolutional patch CNN**, with an architecture inspired by a lightweight patch CNN built for land cover mapping [1], predicting one of **8 classes** (Type I/II chondrule, Matrix, CAIs, Sulfide, Fe oxide, Fe metal, Epoxy) for each pixel from an eight channel patch centered on it. Two 3×3 convolutions and a 1×1 classification head total ≈21k parameters, with no pooling, so **output keeps the same spatial size as the input**:



98.94%

pixel accuracy at the network output



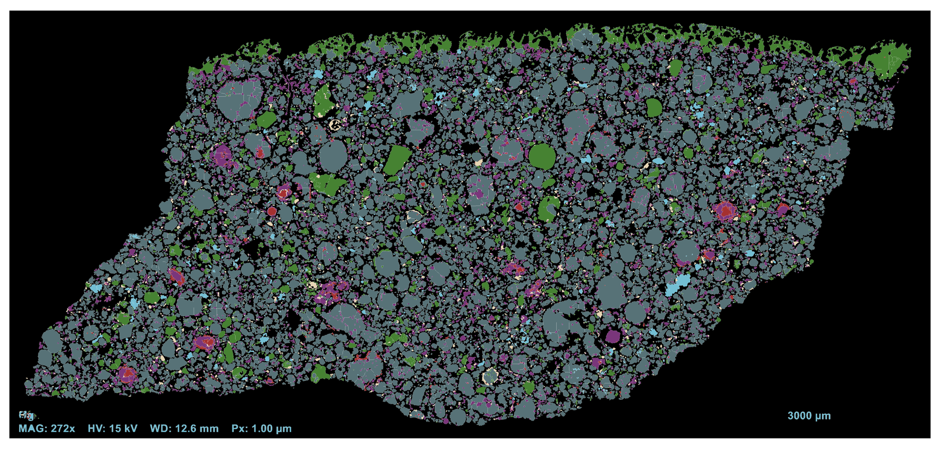
Confusion matrix (colors in log scale)

It was selected by **random search over 100 early and mid fusion configurations** (single 80/20 holdout split), after **screening 25 classical ML classifiers** (about 94% best raw pixel accuracy, with no spatial context, which motivated the patch based CNN).

3. Object extraction

Mathematical morphology cleans the pixel class map (small objects and holes removed, then closing); a **watershed** on the distance transform separates touching grains and draws boundaries between adjacent instances, so **every chondrule, CAI, metal and sulfide grain becomes an individually measurable object**, not just a classified pixel.

A further **region merging** step (region adjacency graph, greedy merge by lowest **cosine distance of mean spectral signatures**, skipping regions already compact (≥ 0.6 fill ratio) or already large (≥ 10,000 px), stopping at $\tau = 0.05$) reduces oversegmentation, but gained little over watershed alone.

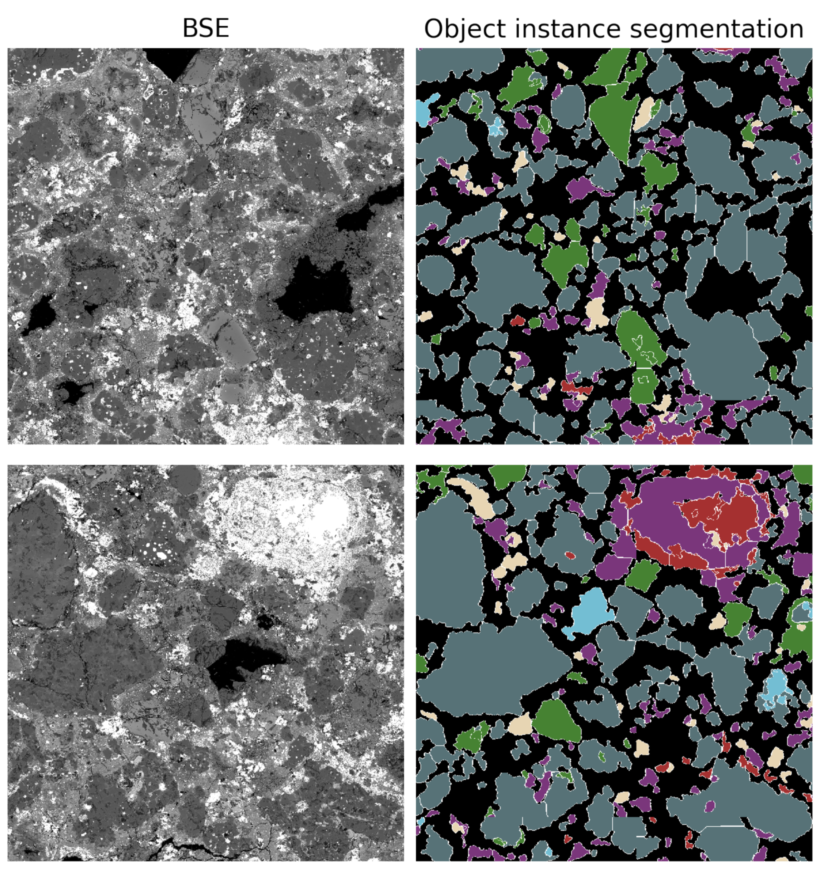


Object instance segmentation over the **full** DOM 08006 section.

4. Morphometry Each object provides its **area, perimeter, eccentricity, solidity and circularity**, computed over the **entire DOM 08006 section** to give the quantitative profile shown above.

Results: Consistency Across DOM 08006

The same CNN and watershed pipeline applied elsewhere on the section, and the resulting identity card for the whole meteorite:



Object instance segmentation at two other locations of DOM 08006, chosen to also show CAIs and Fe grains (oxidized and metallic).

DOM 08006 (CO3.0), whole meteorite identification card

Modal abundance (area %, mineral phases only, excludes epoxy mount)

Class	Area % (of mineral phases)	Objects	Median size (equiv. diam., px)	Median circularity
Type I chondrule	40.4%	6982	41	0.54
Matrix	35.8%	n/a	n/a	n/a
Type II chondrule	11.0%	2182	29	0.43
Fe oxide	7.2%	4495	24	0.54
CAIs	2.2%	628	28	0.54
Sulfide	1.9%	1380	21	0.63
Fe metal	1.4%	765	20	0.59
Epoxy	*39.5% of image*	n/a	n/a	n/a

Whole meteorite identity card: modal abundance, object counts, size and shape, over the **entire DOM 08006 section**.

We Need You: Perspectives & Join Us

- **Proof of concept**: pixels to morphometry, low cost, reproducible.
- Opens quantitative scoring of alteration, metamorphism, shock.
- Next: altered/metamorphosed chondrites, then differentiated objects.

References: [1] Song, Kim & Kim, *Patch-Based Light CNN for Land-Cover Mapping*, Remote Sensing 2019. [2] Griffin *et al.*, *Improved Segmentation of Meteorite Micro-CT Images*, Computers & Geosciences 2012. [3] Weisberg, McCoy & Krot, *Systematics & Evaluation of Meteorite Classification*, 2006.

WE NEED YOU

The real bottleneck is not the model, it is the training set: one chondrite, one annotator, about 48,000 px today. Extending NACHOS needs annotations from domain experts, and **Jonathan Fabrizio** (EPITA LRE) is building a dedicated tool to make that easier.

Annotate, share a dataset, or just talk to us:

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Annotation tool demo video